PHYSCI

Physical Science

Grade 12, Semester 1 study outline

A summary of section headings and key terms, made by students for revision. It is not the handout and does not replace it.

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01. Formation of Elements During Stellar Formation and Evolution

Formation of Elements During Stellar Formation and Evolution

CORE IDEA

- This process of creating elements is called nucleosynthesis.
- The life cycle of stars explains how different elements are produced and scattered across the universe.

Big Bang Nucleosynthesis

BIG BANG NUCLEOSYNTHESIS STELLAR

NUCLEOSYNTHESIS

Happened minutes after the Big

Bang

Happens inside stars

Produced light elements Produces heavier elements

Lasted about 20 minutes Ongoing for billions of years

Key Idea

Life Cycle of a Star

1. Stellar Nebula

2. Protostar

3. Main Sequence Star

- Longest stage of a star's life.

4. Giant Stage

- Red Giant Small to medium stars expand after hydrogen is used up.

5. Final Stage

Important Concepts

- - Stars create the elements needed for planets and life. - Nuclear fusion releases the energy and light of stars. - Gravity helps form stars from nebulae. - Heavier stars create heavier elements. - Many elements in the human body were formed inside stars.

Think About This

1. Why can stars continue shining for millions or billions of years?

2. Why are elements heavier than iron usually formed during supernova explosions?

3. If stars create elements, what does this suggest about the origin of matter on Earth?

Memory Anchor

- "We are made from elements formed inside stars." -Euphe Yireh -- 2 of 2 --

Key terms

Elements heavier than iron - formed during this explosion through neutron capture

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02. Polarity of Molecules

Polarity of Molecules

CORE IDEA

- Atoms do not always share electrons equally.

Electronegativity

- Electronegativity is the ability of an atom to attract shared electrons in a chemical bond.

Important Ideas

- - Higher electronegativity = stronger attraction for electrons - Fluorine (F) is the most electronegative element - Francium (Fr) is one of the least electronegative elements - Valence electrons are the electrons found in the outermost shell

Octet Rule

Bond Polarity

Types of Bond Polarity

EN Difference Bond Type Electron Sharing

- 0 - 0.4 Nonpolar Covalent Equal sharing 0.5 - 2.0 Polar Covalent Unequal sharing 2.1 above Ionic Bond Electron transfer

Key Idea

Examples

Molecule EN Difference Bond Type

N 0 Nonpolar Covalent

Dipole

VSEPR Theory

Main Ideas of VSEPR

1. Electron pairs repel each other

2. Electron pairs stay as far apart as possible

3. Molecular shape depends on bond pairs and lone pairs

4. Lone pairs repel more strongly than bond pairs

Bond Pair vs Lone Pair

Bond Pair Lone Pair

Shared between atoms Not shared

Forms bonds Occupies space around atom

Less repulsion Stronger repulsion

Important Note

Predicting Molecular Shape

Basic Steps

1. Draw the Lewis structure

2. Count bond pairs and lone pairs

3. Use VSEPR geometry

Key terms

Electronegativity - ability of an atom to attract shared electrons in a chemical bond

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03. Intermolecular Forces of Attraction

Intermolecular Forces of Attraction

CORE IDEA

- These forces affect many properties of substances such as boiling point, melting point, solubility, and viscosity.

Intermolecular vs Intramolecular Forces

Intramolecular Forces Intermolecular Forces

Forces within a molecule Forces between molecules

Chemical bonds Molecular attractions

Usually stronger Usually weaker

Important Note

- "INTER" means between molecules.

Three Major Types of IMFA
Type Commonly Found In Relative
Strength
London Dispersion
Forces (LDF)
Nonpolar molecules Weakest
Dipole-Dipole Forces Polar molecules Moderate
Hydrogen Bonding Molecules with H-F, H-O,
H-N
Strongest
1. London Dispersion Forces (LDF)
Key Ideas

- - Electrons move constantly - Temporary dipoles may form -- 1 of 3 -- - These temporary dipoles induce nearby dipoles - Larger molecules usually have stronger LDF Example: O, CH, noble gases

2. Dipole-Dipole Forces
Key Ideas

- - Positive end of one molecule attracts the negative end of another - More polar molecules usually have stronger dipole-dipole attraction Example: HCl molecules attract each other through partial charges.

3. Hydrogen Bonding
Important Note
Properties Affected by IMFA
Property Effect of Stronger IMFA
Melting Point Higher
Boiling Point Higher
Surface Tension Higher
Viscosity Higher

Key terms

Molecules - not only held together internally by chemical bonds. They also attract other molecules through

Hydrogen bonding - stronger than ordinary dipole-dipole forces

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04. Biological Macromolecules

Biological Macromolecules
CORE IDEA

- Living organisms are made of large organic molecules called biomolecules or biological macromolecules.

Biomolecules
Important Terms
Term Meaning
Monomer Small building block molecule
Polymer Large molecule made of repeating
Macromolecule Very large biological molecule
Four Major Biomolecules
Biomolecule Main Function Basic Building Block
Carbohydrates Main energy source Saccharides
Lipids Long-term energy
Proteins Structure and
Amino acids
Nucleic Acids Genetic information Nucleotides
1. Carbohydrates

- Often called saccharides, carbohydrates are the body's main source of energy because they are easily converted into glucose. -- 1 of 3 --

Types of Carbohydrates

Type Description Examples

Monosaccharide One sugar unit Glucose, fructose,

Disaccharide Two sugar units Sucrose, lactose,

Polysaccharide Many sugar units Starch, glycogen,

Important Notes

- - Glycogen stores energy in animals - Starch stores energy in plants - Cellulose gives strength to plant cell walls - Humans cannot digest cellulose completely

2. Lipids

Functions of Lipids

- - Long-term energy storage - Insulation and protection - Formation of cell membranes - Hormone production

Main Types of Lipids

Type Function

Fatty Acids Building blocks of lipids

Phospholipids Main component of cell membrane

Steroids Hormones and membrane stability

Waxes Protection and waterproofing

Key terms

Living organisms - made of large organic molecules called biomolecules or biological macromolecules. These

Biomolecules - large molecules made from smaller building blocks called monomers

Lipids - commonly known as fats. They are hydrophobic, meaning they do not mix with water

Proteins - made of amino acids and are considered important structural and functional molecules in living

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05. Rates of Chemical Reaction

Rates of Chemical Reaction

CORE IDEA

Chemical Reaction

- A chemical reaction is a process where reactants are transformed into new substances called products.

Two Kinds of Change

Physical Change Chemical Change

Changes form only Produces new substances

No new substance formed New substances formed

Usually reversible Often difficult to reverse

Example

- Rusting of iron is a chemical change because a new substance forms.

Collision Theory

- For a chemical reaction to occur, particles must experience an effective collision.

Two Requirements

1. Proper orientation of particles

2. Enough energy during collision

Proper Orientation

- Particles must collide in the correct position for bonds to break and form successfully.

Important Idea

Activation Energy, Ea

- Activation energy is the minimum energy needed for a reaction to occur.

Key Ideas

- - High activation energy slower reaction - Low activation energy faster reaction

Factors Affecting Reaction Rate

Factor Effect on Reaction Rate

Temperature Higher temperature faster reaction

Concentration Higher concentration more collisions

Surface Area Greater surface area faster reaction

Catalyst Lowers activation energy

1. Temperature

Result

Example

2. Concentration

- Higher concentration means more particles are present in the same space.

Result

Example

3. Surface Area

Key terms

Rusting of iron - chemical change because a new substance forms
Activation energy - minimum energy needed for a reaction to occur
Higher concentration - more particles are present in the same space

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06. Limiting Reactants

Limiting Reactants

CORE IDEA

- This reactant is called the limiting reactant because it limits the amount of product that can be formed.

Reactants in a Chemical Reaction

Type Description

Limiting Reactant Reactant consumed first

Excess Reactant Reactant left over after reaction

Important Note

Everyday Analogy

Identifying the Limiting Reactant

General Steps

1. Write the balanced chemical equation

2. Find the stoichiometric mole ratio

3. Determine molar masses

4. Convert grams to moles

5. Find the actual mole ratio

6. Compare actual and stoichiometric ratios

Example

Step 1 - Convert to Moles

Step 2 - Find Ratios

- Stoichiometric ratio: Actual ratio: Since: Aluminum (Al) is the limiting reactant. -- 2 of 3 --

Key Concepts

- - Balanced equations determine mole relationships - Limiting reactants control the amount of product formed - Excess reactants remain after the reaction - Stoichiometry helps predict quantities in reactions

Common Mistakes

- - Forgetting to balance the equation - Comparing grams instead of moles - Mixing up limiting and excess reactants - Using incorrect molar masses

Think About This

1. Why must grams be converted to moles before comparing reactants?
2. Why does the reaction stop once the limiting reactant is used up?
3. Can two reactants both be limiting at the same time?
Memory Anchor

Key terms

Once the limiting reactant - completely used up, the reaction stops even if other reactants are still available

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07. Sources of Energy

Sources of Energy
PHYSICAL SCIENCE - LESSON 7
SOURCES OF ENERGY
CLASS, LET'S BEGIN..
GUIDING QUESTION
WHY IT MATTERS
LET'S BUILD THE IDEA
THE COMMON CHAIN
1. Start with the source: heat ba, moving water, moving air, or another usable form of energy?
D. Finally, the electrical energy is transmitted for use in homes, schools, and communities
SIR'S IMPORTANT EXCEPTION
RENEWABLE
ENERGY SOURCES

- The Agus- Pulangi system is the lesson's Mindanao example.
- Kapag walang useful wind, walang rotation, and therefore, walang electrical output.
- Pero hindi automatic na sustainable dahil organic: the material must be replaced and managed responsibly.

DON'T MIX THESE UP
PATTERN CHECK
NON-RENEWABLE
ENERGY SOURCES
ENVIRONMENT CHECK
SAME MACHINE, DIFFERENT STARTER
COMPARE WITH SENSE

- Real energy decisions involve tradeoffs, so compare the sources using these lenses: a.

STEWARDSHIP CHECK
USE IT
SIR'S PRACTICAL CHECK
1. Name the source first. Renewable ba or non-renewable?
3. Hanapin ang turbine and generator, or explain why the system does not need them
4. Give one practical benefit and one limitation or consequence. Dapat parehong meron
5. Final check: naipaliwanag mo ba ang process, or pinangalanan mo lang ang source?
YOUR TURN
SIR'S NUDGE
THINK ABOUT IT

Key terms

Renewable sources - naturally replenished and can be used repeatedly. Sa madaling sabi, napapalitan ang
Pulangi system - the lesson's Mindanao example

08. Active Ingredients of Cleaning Products

Active Ingredients of Cleaning Products

PHYSICAL SCIENCE - LESSON 8

ACTIVE INGREDIENTS

OF CLEANING PRODUCTS

CLASS, LET'S BEGIN..

GUIDING QUESTION

WHY IT MATTERS

- Kaya the smartest cleaner is not automatically the strongest one, it is the correct one, used correctly.

LET'S BUILD THE IDEA

CORE IDEA

- Cleaning is not always disinfecting.

SAFETY BEFORE CHEMISTRY

- Never mix bleach or disinfectants with other cleaners unless the labels explicitly say it is safe.

THE SEVEN ACTIVE INGREDIENTS

SIR'S CHECK

MECHANISM CHECK

- Never combine an acid cleaner with bleach or another product unless the labels explicitly permit it.

DON'T GENERALIZE

LIKE DISSOLVES LIKE

- Examples in the lesson include sodium hypochlorite, phenolics, and quaternary ammonium compounds.

LANGUAGE CHECK

- Pero the name alone is never enough.

SIR'S COMPARISON LENS

Kapag nagko-compare ng cleaners,

FORMULATION CHECK

- Kaya identify each role, then let the label, not your guess, decide how the complete product should be used.

USE THE LABEL, NOT THE GUESS

MATCH THE INGREDIENT TO THE JOB

SIR'S WARNING

- This is a reasoning guide, not permission to experiment with mixtures.
- The actual product label always has the final say.

USE IT

SIR'S LABEL-READING ROUTINE

1. Define the job: grease ba, mineral deposit, stain, visible dirt, or germs?

2. Identify the active ingredient or ingredient family on the label

3. Confirm the approved surface and intended use. Safe ba ito for the material you will clean?

4. Read dilution, application, contact-time, ventilation, and protective-equipment directions

6. After use, close, store, and dispose of the product exactly as directed

YOUR TURN

Key terms

Water - polar; many oily substances are non-polar. Kaya an appropriate organic solvent can dissolve residues that water

Disinfectants - intended for nonliving surfaces. Products used safely on skin or wounds are more accurately called

09. Ingredients in Cleaning Agents

Ingredients in Cleaning Agents

I. 4 Types Of Cleaning Agent Ingredients

1. Functional Cleaning Agents

Cleaners/Whiteners with strong actions

Ingredient Main Function Examples

SILICATES Boosts cleaning power, protects

Laundry detergent booster

AMMONIA Cuts through grease, evaporates

A. Sodium bicarbonate (Baking Soda)

- Sodium bicarbonate (NaHCO), commonly known as baking soda, is a mild, natural cleaning agent.

B. Silicates

C. Ammonia

- Ammonia (NH) is a powerful, fast-evaporating cleaning agent known for cutting through grease, grime, and stains.

D. Bleaching Agents

2. Biological Helpers

Ingredient Main Function Examples

A. Enzymes

Lipases - break down fat and grease

Amylases - break down carbohydrates/starches

3. Stability & Texture Additives

Ingredient Main Function Examples

PRESERVATIVES Prevents spoilage by

A. Polymers

B. Processing Aids

C. Preservatives

4. Aesthetic Additives

Ingredient Main Function Examples

- -- 3 of 4 -- COLORANTS Adds color for appeal or coding Gel-based cleaners, shampoo

FRAGRANCES Adds pleasant scent Emulsifiers in cleaning pastes

A. Colorants

They provide a product with an individual

B. Fragrances

- 10 Household Products and their Active Ingredients.
- Retrieved from Sporcle: https://www.sporcle.com/games/trumpeteerx/common-household-chemicals Types of Household Cleaning Products. (n.d.).

Key terms

Bleaching agents - chemicals used to whiten, disinfect, and remove stains by breaking down colored
Enzymes - biological molecules (proteins) used in cleaning products to break down specific stains and
Polymers in cleaning agents - large molecules made up of repeating units that help trap dirt, prevent re-
Processing aids - ingredients added to cleaning products to enhance manufacturing, improve stability, or
Preservatives - chemicals added to cleaning products to prevent the growth of bacteria, mold, and fungi
Fragrances - added to cleaning products to give them a pleasant smell, mask chemical odors, and enhance

10. Ancient to Modern Astronomy

Ancient to Modern Astronomy

I. The Greeks and the Spherical Earth

- The ancient Greeks were among the first to suggest that Earth is a sphere-long before modern technology confirmed it.

Clues that Earth is Spherical

1. Ship's Hull Disappearing First

- - As ships sailed away, the hull vanished before the sails. - This wouldn't happen if Earth were flat. - The gradual disappearance is explained by a curved surface.

2. Phases of the Moon & Lunar Eclipses

- - The shadow Earth casts on the Moon during a lunar eclipse is always round. - Only a spherical object can cast a consistently round shadow. -- 1 of 7 --

3. Eratosthenes' Measurement of Earth's Size (c. 240 BCE)

II. Ancient Astronomical Phenomena

Planetary Movement

- - "Planet" comes from the Greek word plants, meaning "wanderer." - Unlike stars, planets move across the night sky. - Ancient observers noted that planets sometimes moved backward in the sky, a puzzling motion called retrograde.

Retrograde Motion of Mars (Ptolemaic Model)

- 150 AD) proposed the geocentric model, where Earth is at the center of the universe. - To explain Mars' retrograde motion, he introduced epicycles-small circular orbits on top of larger orbits. - Mars appeared to move backward when it looped along an epicycle. -- 3 of 7 --

Copernican Model (Heliocentric Theory)

- Nicolaus Copernicus proposed a heliocentric model-the Sun is at the center, and Earth & planets orbit it. - Retrograde motion was naturally explained: - Earth overtakes Mars in orbit, making Mars appear to move backward temporarily. - This was a major shift in understanding-later supported by Galileo and Kepler.

Solar and Lunar Eclipses

- - Solar Eclipse: Occurs when the Moon blocks the Sun's light and casts a shadow on Earth. - Lunar Eclipse: Happens when Earth casts a shadow on the Moon during a full moon. - Ancient people often saw eclipses as omens, though many civilizations could predict them accurately.

North Star (Polaris)

- - Located nearly above Earth's North Pole, Polaris stays almost fixed in the night sky. - Used by ancient navigators and explorers to determine direction (true north). - Its unchanging position helped prove Earth spins on an axis. -- 4 of 7 --

III. Laws Of Planetary Motion

Tycho Brahe (1546-1601)

- - A Danish nobleman and astronomer known for his extremely accurate naked-eye observations. - Built advanced instruments to measure the positions of stars and planets. - Did not support the heliocentric model but collected data that helped prove it. - His decades of data, especially on Mars, were essential for Keple...

Johannes Kepler (1571-1630)

- - A German mathematician and assistant to Tycho Brahe. - Used Tycho's planetary data to develop 3 laws of planetary motion that support the heliocentric model.

Kepler's Three Laws of Planetary Motion

1. Law of Ellipses

- - Planets orbit the Sun in an ellipse, with the Sun at one focus. - This corrected the belief that orbits are perfect circles.

2. Law of Equal Areas

- - A line from a planet to the Sun sweeps out equal areas in equal times. - Planets move faster when closer to the Sun and slower when farther. -- 5 of 7 --

3. Law of Harmonies

- - The square of a planet's orbital period is proportional to the cube of its average distance from the Sun. - Explains how planets farther from the Sun take longer to orbit.

Diliman, Quezon City

11. Aristotelian and Galilean Concept of Motion

Aristotelian and Galilean Concept of Motion

Motion

I. Nature of Motion

Aristotelian Concepts on Motion

Galilean Concepts on Motion

- - "Motion is a natural state which continues until altered" - motion continued unless interfered with. - force as interference of motion rather than cause. - without gravity to speed the descent and slow the ascent, and without friction, objects would not start or stop moving at all. - discovery of the acceleration ...

II. Vertical Motion

Aristotle

- - the behavior of objects depends upon their composition - heavy objects, like stones, tend to fall downwards.

Galileo

- - pure freefall acceleration is uniform and constant for all objects regardless of their own size and weight, and that acceleration is 9.8 m/s2 - first established the idea that all objects on the surface of the earth are being pulled by the earth's gravitational force at an acceleration of 9.8 m/s2, so any object t...

III. Projectile Motion

Aristotle

Galileo

- - the motion of a projectile is a combination of constant horizontal velocity and vertical motion, in which the projectile accelerates at a rate of 9.8 m s2

IV. Acceleration in a Vacuum

- It was mentioned earlier that friction is an interference to motion.
- This means the presence of friction can slow down or stop motion.
- The above observations were made by Galileo during his time which led him to conclude that all objects fall with the uniform acceleration in vacuum.
- A vacuum is a place where there is no air, therefore, no air resistance.

V. Difference Between Galileo's Assertion of Frictionless Motion and Newton's Law of Inertia

- What is the difference between the two ideas?
- It is the terminology used by the two scientists.
- Galilean Forms of Explanation(Retrieved October 24, 2020 from http://www.drabruzzi.com/aristotelian_vs_galileian.htm Canoy, Warlito Z.

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12. Propagation of Light, Reflection and Refraction

Propagation of Light, Reflection and Refraction

General Behavior of Light

- Reflection: The bouncing of light when it reaches a reflective surface or boundary between two media. - Wave Model: Similar to water and sound waves, light waves reflect off surfaces at the same angle they hit (Law of Reflection). - Particle Model: Photons (particles of light) bounce off surfaces like tiny balls, re...
- The angle of incidence () between the incident ray and the normal (CM) is equal to the angle of reflection () between the reflected ray and the normal.

Two Types of Reflection

- This type of reflection allows you to see clear images in mirrors.
- This type of reflection enables us to see most objects around us. -- 2 of 6 -- Refraction is the bending of light when it passes obliquely from one medium to another of different density or through different layers within the same medium.

Laws of Refraction

- Index of Refraction: The refractive index () is the ratio of the speed of light in a vacuum () to the speed of light in the material (): = The refractive index is always greater than 1. -- 3 of 6 --

Dual Nature of Light

Particle Model (Isaac Newton)

- - Light consists of particles (corpuscles) that travel in straight lines. - Reflection: Photons bounce off surfaces, similar to how balls bounce. - Refraction: Photons change direction due to a force at the boundary between two media.

Wave Model (Christiaan Huygens)

Red Light in Photographic Dark Rooms

Sunburn from Ultraviolet (UV) Light

Perception of Colors

- 60-70) Ariew, Roger et al.

Key terms

Reflection and Refraction - fundamental behaviors of light that can be explained using both the wave and

Refraction - bending of light when it passes obliquely from one medium to another of different density or

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13. Wave Properties of Light

Wave Properties of Light

- Topic: Wave Properties of Light

I. Experiment showing that electrons behave like waves

- Light exhibits both wave-like and particle-like properties (this is called wave-particle duality).

Davisson and Germer Experiment

II. Wave Properties of Light

1. Dispersion of Light

- It was Isaac Newton who first discovered that ordinary white light is a combination of colors.

Formation of Rainbow

2. Scattering of Light

A. Rayleigh Scattering

B. Mie Scattering

C. Non Selective Scattering

3. Interference

4. Diffraction

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14. Various Light Phenomena

Various Light Phenomena

A. Your reflection on the concave and convex sides of a spoon looks different

- The bulging side is the convex side, where the angle of incidence equals the angle of reflection, but the surface normal changes across points.
- Convex mirrors make images smaller, especially near the edges, and are used in traffic and stores to see around corners or spot shoplifters.
- Turn the spoon to the side that curves inward-this is the concave side, like a cave.

B. Mirages

C. Haloes, sundogs, primary rainbows, secondary rainbows, and supernumerary bows

Sundogs form as sunlight is refracted by hexagonal

D. Why are clouds usually white and rain clouds dark?

E. Why is the sky blue and sunsets are reddish?

Key terms

Supernumerary bows - faint, colorful bands inside the primary rainbow, caused by interference of light. They

15. Special theory of Relativity

Special theory of Relativity

I. Newtonian Mechanics Vs Maxwell's Electromagnetic Theory

- The propagation speed v electromagnetic waves in a vacuum is calculated using the formula: This value is the speed of light in a vacuum, showing that light is an electromagnetic wave. "speed of light is constant" This is where the conflict between Newtonian mechanics and Maxwell's theory starts.
- Newtonian mechanics couldn't explain this, because it assumed that velocities simply add - which would mean different observers should measure different speeds for light.

II. Special Theory of Relativity

1. Postulates of Special Theory of Relativity

A. Postulate #1:Principle of Relativity

- The laws of Physics are the same in all inertial frames of reference moving with constant velocity relative to one another

B. Postulate #2: Constancy of the Speed of Light

- The speed of light is the same in all inertial frames of reference

2. Consequences of Special Relativity

A. Time Dilation

B. Length Contraction

- This means long interstellar journeys could seem shorter to travelers, even though they still take years or centuries in Earth time.
- Twin Paradox: The Twin Paradox is a thought experiment in special relativity where one twin travels at near light speed in space while the other stays on Earth.

C. Mass - Energy Equivalence

- E = mc2 Mass and energy are two forms of the same thing.
- This means that mass and energy are equivalent.
- Interpretation: - Mass and energy are two forms of the same thing. - A small amount of mass can be converted into a huge amount of energy, because c2 is a very large number. - No massive object can ever reach or surpass the speed of light Dasas, Melody D. and Villavert, Joam C.

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16. Postulates of General Relativity

Postulates of General Relativity

Consequences of General Relativity

- What Einstein proposed was a new theory of gravity, in which mass determines the curvature of spacetime and that curvature, in turn, controls how objects move.

Figure 1 Newtonian Gravity vs General Relativity

The Motion of Mercury

Figure 2 Mercury's Wobble

- -- 1 of 4 -- According to Newtonian gravitation, the gravitational forces exerted by the planets will cause Mercury's perihelion to advance by about 531 seconds of arc (arcsec) per century.
- But no planet has ever been found nearer to the Sun than Mercury, and the discrepancy was still bothering astronomers General relativity, however, predicts that due to the curvature of spacetime around the Sun, the perihelion of Mercury should advance slightly more than is predicted by Newtonian gravity.
- The prediction of general relativity is that the direction of perihelion should change by an additional 43 arcsec per century.

Deflection of Starlight

- Since space time is more curved in regions where the gravitational field is strong, we would expect light passing very near the Sun to appear to follow a curved path, Einstein calculated from general relativity theory that starlight just grazing the Sun's surface should be deflected by an angle of 1.75 arcsec.
- Starlight passing near the Sun is deflected slightly by the "warping" of space time. (This deflection of starlight is one small example of a phenomenon called gravitational lensing.) Before passing by the Sun, the light from the star was

traveling parallel to the bottom edge of the figure.

Black Hole

- The most fundamental prediction of general relativity is the existence of black holes.
- A black hole is a place in space where gravity pulls so much that even light cannot get out.
- Mass is the amount of matter, or "stuff," in an object.
- Another kind of black hole is called "stellar."

Figure 5 Blackhole

- Earth's galaxy is called the Milky Way.
- The largest black holes are called "supermassive." These black holes have masses that are more than 1 million suns together.
- The supermassive black hole at the center of the Milky Way galaxy is called Sagittarius A.
- It has a mass equal to about 4 million suns and would fit inside a very large ball that could hold a few million Earths.
- When a black hole and a star are close together, high- energy light is made.
- This kind of light cannot be seen with human eyes.

Figure 4 Curvature of Light Paths near the Sun

Key terms

Since space time - more curved in regions where the gravitational field is strong, we would expect light passing

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17. Speeds and Distances of Far-off Objects

Speeds and Distances of Far-off Objects

Celestial Mechanics

Cosmic Distance Ladder

- This method is composed of several methods that are built on one another.
- The data obtained in the first step of the ladder are used in the succeeding steps and so on.
- The base of the ladder is a measurement without any assumptions about stars' characteristics.
- For example, the measurement of one astronomical unit (AU), which is the measure of the distance of Earth from the Sun, is considered the base.
- This value is used in measuring the parallax of a star.

Parallax

- Parallax is the apparent change in the position of an object due to change in the way it is perceived, depending on the perception of the viewer.
- It is used in measuring the distance of the stars that are approximately 300 light years away.

Remember that,

- -- 1 of 5 -- 1 degree = 1/360 degree of a circle 1 arc second= 1/3600 of a degree We, can establish the relationship between a star's distance and its parallax angle as: Usually, this distance is expressed in units of parsec.
- Parsec (parallax second) is the distance of a star that has a parallax of two arcseconds.
- Arcsecond is the 60th part of one arcminute and one arcminute is the 60th part of one degree.
- But, a more common unit used to denote distances of celestial objects is light year.
- A light year is defined as a distance light can travel in one year.
- The conversion factor is presented below: One light year= 0.306601 PARSEC, which is equal to 9.461 x 1015 m

Spectroscopic Method

- For stars whose parallax cannot be measured using the ladder, the spectroscopic method is used.
- The standard measure of the brightness of a star in astronomy is the brightness of a star that is 10 parsec away from earth.
- This brightness is called absolute brightness while the actual brightness of the star that we see here on earth is known as its apparent brightness.
- By looking at their spectral lines under the process known as spectroscopy, astronomers analyze the spectra of nearby stars whose parallax are known to those which are not.
- Astronomers are able to determine the spectral type of a star's spectrum by analyzing its spectral lines and plotting the observations in the Hertzsprung - Russell diagram.
- Hertzsprung-Russell (HR) Diagram is a graph that shows a star's luminosity versus its temperature.
- It is an important tool in determining the distance of far-off objects because astronomers believe that the stars near

Earth are similar to the stars far from earth.

Doppler Effect

- It is the shift in the wavelength of the emitted light of an object which is proportional to the speed with which the object moves.
- All galaxies outside our galaxy indicate a red shift, moving away from us. -- 2 of 5 -- where v is the speed of the object c is the speed of light is the rest wavelength, and is the measured wavelength.

By substituting the values,

Distances up tp 13,000,000 light years

- If somebody obtains the distance to a given galaxy, Cepheid variables must be located first.

Distances up to 1,000,000,000 light years

- Supernova is a star that usually blows itself apart at the end of its life and becomes so bright for a period of time.
- To measure the distance, the distance of the supernova to Earth must be known.
- Then the periodic brightness and dim must be obtained.

Distances beyond 1,000,000,000 light years

- For every far object, the General Theory of Relativity must be used to measure distances.
- Indeed the cosmic ladder distance can be used to estimate the distances of objects in space and it also allowed astronomers to reach the conclusion that our universe is composed of vast collections of galaxies.

Cosmic Distance Ladder

Key terms

Parallax - apparent change in the position of an object due to change in the way it is perceived, depending on

Cepheid stars - very abundant and bright in the galaxy. If somebody obtains the distance to a given galaxy

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18. The Expanding Universe

The Expanding Universe

The Universe as Perceived by Early Scientists

- These scientists include Heinrich Olbers, Sir Isaac Newton, and Albert Einstein.

Heinrich Olbers

- a German physician and astronomer, argued that if the universe is infinite, we should be seeing a night full of stars having no part of darkness.

Sir Isaac Newton

- an English physicist and mathematician, introduced the concept of gravity-a force of attraction, and argued that if the universe is finite, it should be collapsing on itself due to the attractive force between objects within the universe.

Albert Einstein

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- This phenomenon is known as the Doppler Effect. -- 1 of 4 -- Figure 1: Doppler Effect The Doppler Effect is evident when you hear the changing pitch of an ambulance siren as it passes you.
- The Doppler Effect is also observed in light.
- When a light source approaches, there is an increase in its measured frequency.
- An increase in frequency is called blueshift because the increase is toward the high-frequency or blue end of the color spectrum.
- A red shift refers to a shift toward the lower-frequency or red end of the color spectrum of distant galaxies.
- If all the galaxies surrounding us move away from our galaxy does it mean we are the center of the universe?
- This means that the universe has no center and is expanding in all directions.

1. The open universe that tends to expand without end

Evidence of the Big Bang Theory

1. Cosmic Microwave Background Radiation (CMB)

- Explanation: The CMB is the thermal radiation left over from the time of recombination, approximately 380,000 years after the Big Bang.

2. Expansion of the Universe

3. Abundance of Light Elements

4. Large-Scale Structure of the Universe

5. Redshift of Galaxies

6. Primordial Gravitational Waves (Theoretical)

Key terms

H - the Hubble's constant equal to =